

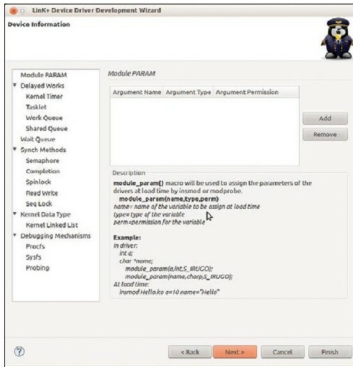


Figure 4: Linux kernel features

window options change dynamically.

Click on **Next** for specific information on the driver and device, or on the **Finish** button to create the project with the above features.

The third (and last) page has Linux device driver specific features, and is divided into two parts as shown in Figure 5.

The first half contains the types of drivers, devices and the bus type, as well as the framework method.

- In the **Driver Type** field, select the name of the driver from the list.
- In the **Bus Type** field, select the bus name from the list.
- In the **Device Type** field, select the device's name from the list.
- In the **Framework Method** field, select the name of the framework method from the list.

The second half is dynamic and depends on the user's first selection. It contains the following:

- **File operations:** This has a list of functions. You can select zero or more functions from the list.
- **Registration view:** This depends on what you selected in the first half.

Basic registration tab fields are compulsory and other registration fields like **IRQ**, **IOCTL**, **Port** and **Buffer Management** are optional. Finally, click on **Finish** to generate the complete and specific code for the device driver.

## Typical driver development

You have an option to choose **Typical Driver Development** in the **General Information** page. Select it and click **Next**. This opens a wizard as shown in Figure 6.

Unlike in *Custom driver development*, in *Typical driver development* you are asked only driver specific information. Based on your requirement, choose the **Framework types**,

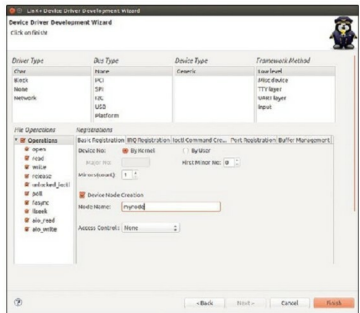


Figure 5: Linux device driver specific features



Figure 6: Driver development wizard

**Driver types** and **Bus infrastructure** from the left pane either by double-clicking on the specific type, or by dragging and dropping on to the right pane. Once this job is done, click **Finish** to generate the device driver code. This in turn speeds up and eases the job of a device driver developer.

## Project build and utility options

By default, **Build Automatically** is enabled and if you want to disable this feature, select **Project** and uncheck **Build Automatically**. To build a project, right-click on that project and then click on **Build Project**.

The Link+ IDE is bundled with different device driver utility options like **insmod**, **rmmod**, **modprobe**, **modinfo**, **dmesg** and **lsmod**. These options can be accessed in multiple ways. One way is to right click on the **Project** and in the **Project Explorer** view, select **Link+ Driver Development**, as